

Epistemic Case Study — Findings, with Data

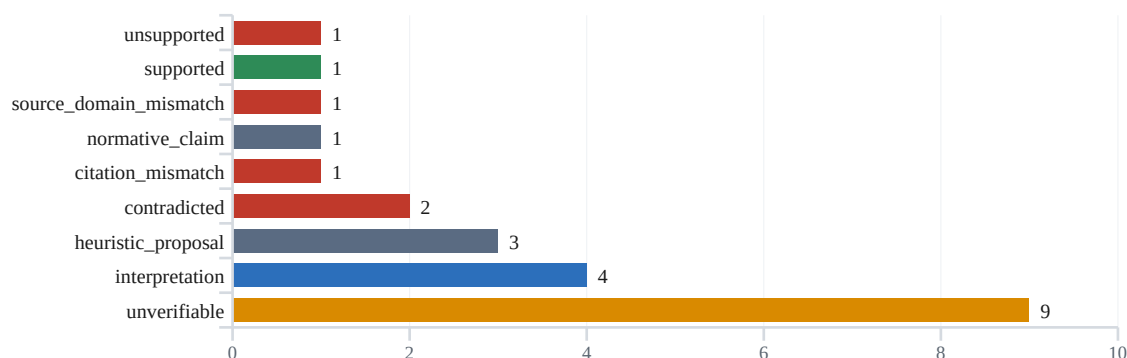
The “epistemic validation” of a Muse Spark 1.1 text by MarCognition-AI, analysed with DESi

Source: Hugging Face post “*Epistemic Stress Test — Muse Spark 1.1 validated by MarCognition-AI*” (user elly99, 2026-07-10; Zenodo 10.5281/zenodo.20509721) and github.com/elly99-AI/MarCognition-AI. Regenerated offline & deterministically with `python -m desi.case_studies.marcognition_muse_spark`. Every number below comes from `summary.json` / `claims.jsonl` / `evidence.jsonl`.

23 atomic claims typed & anchored	4/23 claims with domain- admissible evidence	3 structural contradictions (detector)	yes self-sealing conclusion; falsifier: no
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Distribution of verdicts (23 claims)

No binary VERIFIED/FAILURE stamps: heuristics, interpretations and normative statements get their own categories; “unverifiable” is a first-class verdict, not a fallback.



Evidence strength — where concrete support is missing



18 of 23 claims have no concrete evidence passage at all — the validator referred throughout to “the PubMed document” with no title or locus.

The three structural contradictions

Found by DESi's own detector `desi.self_audit.contradictions.find_contradictions` — not asserted by prose.

C1 — Prompt ↔ Method

The Method (muse:L206): “No epistemic instructions (verification, sources, stages)”. The printed prompt demands ≥5 references with direct citations (L56–58), a citation-consistency check (L64) and six named Phases (L29–47). The experiment contradicts its own setup.

C2 — “all VERIFIED” ↔ “no citations”

The same report stamps five claims VERIFIED (L170–198) and ends with “No citations found or verifiable” (L201–202) — although the text carries eight references (L154–161). Two decoupled subsystems, concatenated without reconciliation (agent_metacognition L48–66).

C3 — “independent” ↔ one LLM call

“Independent external validator” (L208) vs. the implementation: a single `llm.invoke` (skeptical_agent L62) that receives “the reference documents used for generation” (evaluator_prompt L24–28).

Self-sealing & falsifiability

Would support	Validator flags the unverifiable → the gap is made visible; OR the validator fails → the Boundary “within the validator itself” (L237). Both outcomes confirm.
Would weaken	A pre-registered run in which the validator, given domain-correct sources, confirms claims with quoted passages — not provided.
Would refute	A control in which the residual failures vanish under proper source-gating (a pipeline defect, not an “intrinsic architecture”) — not provided.
Falsification condition stated?	No. Since success and failure both confirm and no outcome is designated disconfirming, the hypothesis is unfalsifiable as <i>run</i> .

All 23 claims — the data

Complete and line-by-line auditable in `claims.jsonl` / `evidence.jsonl`. Column “Evidence” = provenance kind of the source actually available.

ID	Type	Domain	Verdict	Evidence
MET-01	methodological	experiment methodology	contradicted	primary
MET-02	methodological	experiment methodology	contradicted	primary
MET-03	methodological	experiment methodology	unverifiable	none
VAL-01	claim about validator	legal philosophy	source_domain_mismatch	semantic-only
VAL-02	claim about validator	experiment methodology	citation_mismatch	primary
VAL-03	claim about validator	experiment methodology	supported	primary
EB-01	claim about epistemic boundary	ml epistemology	interpretation	none
EB-02	claim about epistemic boundary	ml epistemology	unsupported	secondary
DEF-01	definition	legal philosophy	interpretation	none
DEF-02	definition	legal philosophy	interpretation	none
ATTR-01	author attribution	legal philosophy	unverifiable	none
ATTR-02	author attribution	legal philosophy	unverifiable	none
QUOTE-01	direct quote	legal philosophy	unverifiable	none
QUOTE-02	direct quote	legal philosophy	unverifiable	none
FACT-01	historical fact	legal philosophy	unverifiable	none
HEUR-01	heuristic model	legal philosophy	heuristic_proposal	none
HEUR-02	heuristic model	legal philosophy	heuristic_proposal	none
HEUR-03	heuristic model	legal philosophy	heuristic_proposal	none
THEO-01	theoretical interpretation	economics	interpretation	none
EMP-01	empirical	positive law	unverifiable	none
FACT-02	juristic fact	positive law	unverifiable	none
FACT-03	historical fact	technology history	unverifiable	none
NORM-01	normative	legal philosophy	normative_claim	none

MarCognity vs. DESi — what differs

Not a sales pitch: DESi does not prevent errors, it makes visible the point at which an error, an omission or an inadmissible inference arises.

Dimension	MarCognity (Skeptical Agent)	DESi
Claim coverage	5 general claims	23 typed, incl. quotes/attributions
Source fit	none (PubMed ↔ law)	source_domain_gate → mismatch, not VERIFIED
Concrete provenance	“the PubMed document”	exact anchor (doc:line) or none
Contradiction detection	misses C1, produces C2	C1/C2/C3 via find_contradictions
Interpretation/heuristic	binary VERIFIED/FAILURE	heuristic_proposal / interpretation / normative
Uncertainty	global prose	per claim: verdict + uncertainty + strength
Falsifiability	no condition	names support/weaken/refute + missing falsifier
Auditability	one concatenated text	jsonl per line + optional hash-ledger
Evaluator self-check	none; failure = confirmation	the report is itself under test (VAL-01..03)

Core finding

The demonstration is not that Muse Spark makes mistakes. It is: the claimed validation **confirms general legal statements with unsuitable, non-transparent sources** (PubMed for legal philosophy, without title/passage), **overlooks a direct contradiction in the setup (C1)** and then **reads its own failure as confirmation of the theory** (self-sealing, with no falsification condition).

Limits of DESi (honestly)

DESi is not infallible: the claim fixture is curated (no auto-extractor); the legal philosophy is *not* adjudicated on the merits here (many claims deliberately end on “unverifiable”); source_domain_gate and the self-sealing analysis are small, general extensions. MarCognity's own README/Boundary document are more careful than the forum conclusion — the over-reach is in the conclusion, not in the hedged hypothesis.